

Chapter 5

Extensions of the Least-squares Method

5.2 Output-error method

Denote the true process as

$$y(t) = \frac{B^o(q)}{A^o(q)} u(t) + v(t)$$

and the identified model as

$$G(q) = \frac{B(q)}{A(q)} = \frac{b_1 q^{-1} + \dots + b_n q^{-n}}{1 + a_1 q^{-1} + \dots + a_n q^{-n}}$$

Parameter vector

$$\theta = \begin{bmatrix} a_1 \\ \vdots \\ a_n \\ b_1 \\ \vdots \\ b_n \end{bmatrix}$$

5.2 Output-error method

Given the data set

$$Z^N := y(1) u(1) \cdots \cdots y(N) u(N)$$

The predictor is

$$\hat{y}(t | \theta) = \frac{B(q)}{A(q)} u(t)$$

Introduce the output error

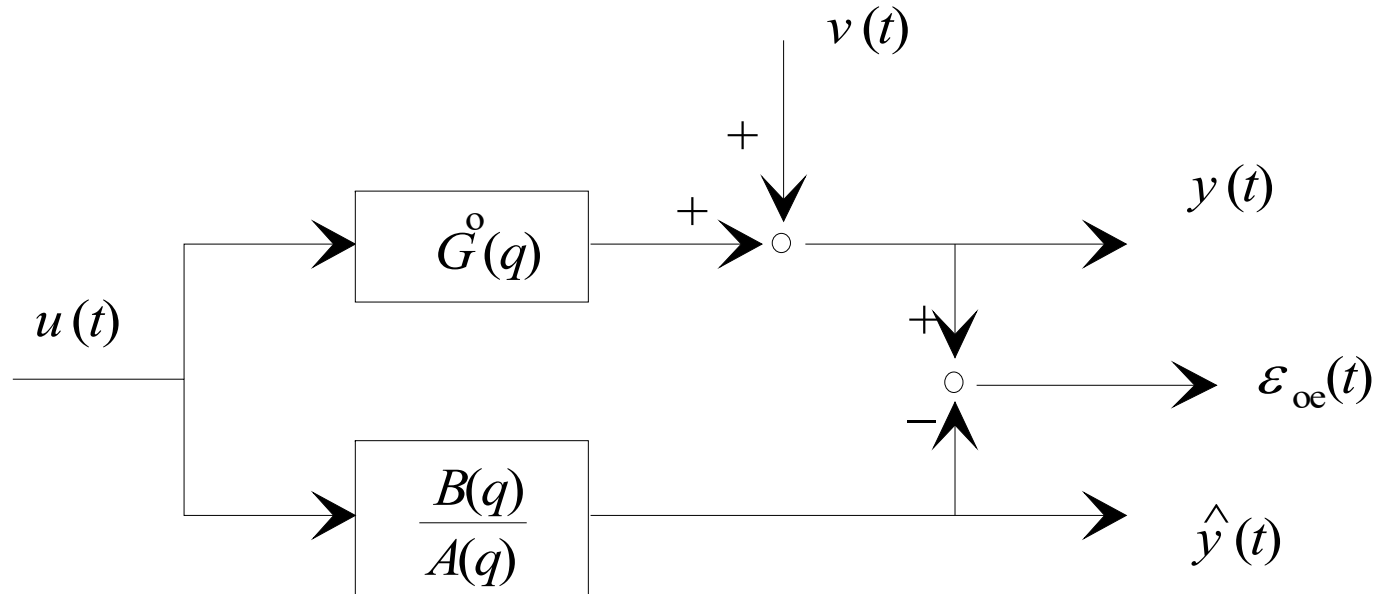
$$\varepsilon_{\text{oe}}(t) = y(t) - \hat{y}(t) = y(t) - \frac{B(q)}{A(q)} u(t)$$

Estimate the parameter vector $\hat{\theta}$ by minimizing

$$V_{\text{oe}}^N = \frac{1}{N} \sum_{t=1}^N \varepsilon_{\text{oe}}(t)^2$$

5.2 Output-error method

Block diagram



A big numerical problem: The output error is nonlinear in $A(q)$ parameters. This is a *nonlinear* least-squares problem which needs an iterative optimization routine and is much more complex problem than the ARX model estimation.

5.2 Output-error method

Analysis: Proof of consistency

Assumptions:

- 1) The true process is given by (5.1.4) where $v(t)$ is stationary stochastic process with zero mean.
- 2) The model order n is correct.
- 3) The input is persistently exciting with order higher than $2n$.
- 4) The test is in open loop.**
- 5) The minimization converges to the global minimum for all N .
- 6) $A(q)$ and $B(q)$ have no common factor.

5.2 Output-error method

Analysis: Proof of consistency(asymptotically unbiasedness).

Denote the model error as

$$\Delta G(q) = G^o(q) - \hat{G}(q)$$

When $N \rightarrow \infty$

$$\begin{aligned} V_{\text{oe}}^N &\rightarrow EV_{\text{oe}} = E[\Delta G(q)u(t) + v(t)]^2 \\ &= [\Delta G(q)]^2 Eu^2(t) + 2\Delta G(q)E[u(t)v(t)] + Ev^2(t) \\ &= [\Delta G(q)]^2 Eu^2(t) + Ev^2(t) \end{aligned}$$

When the minimization converged to its global minimum, we have

$$[\Delta G(q)]^2 \rightarrow 0 \quad \text{as } N \rightarrow \infty$$

5.2 Output-error method

Analysis: Proof of consistency (asymptotically unbiasedness).

Because model parameters are uniquely determined (identifiable), this is equivalent to saying that

$$\hat{\theta} \rightarrow \theta^o \text{ w.p. 1 as } N \rightarrow \infty$$

This is the consistence of the estimate.

5.2 Output-error method

Simulation Example: Process

$$y(t) = \frac{0.5q^{-1}}{1 - 0.97q^{-1}} u(t) + v(t)$$

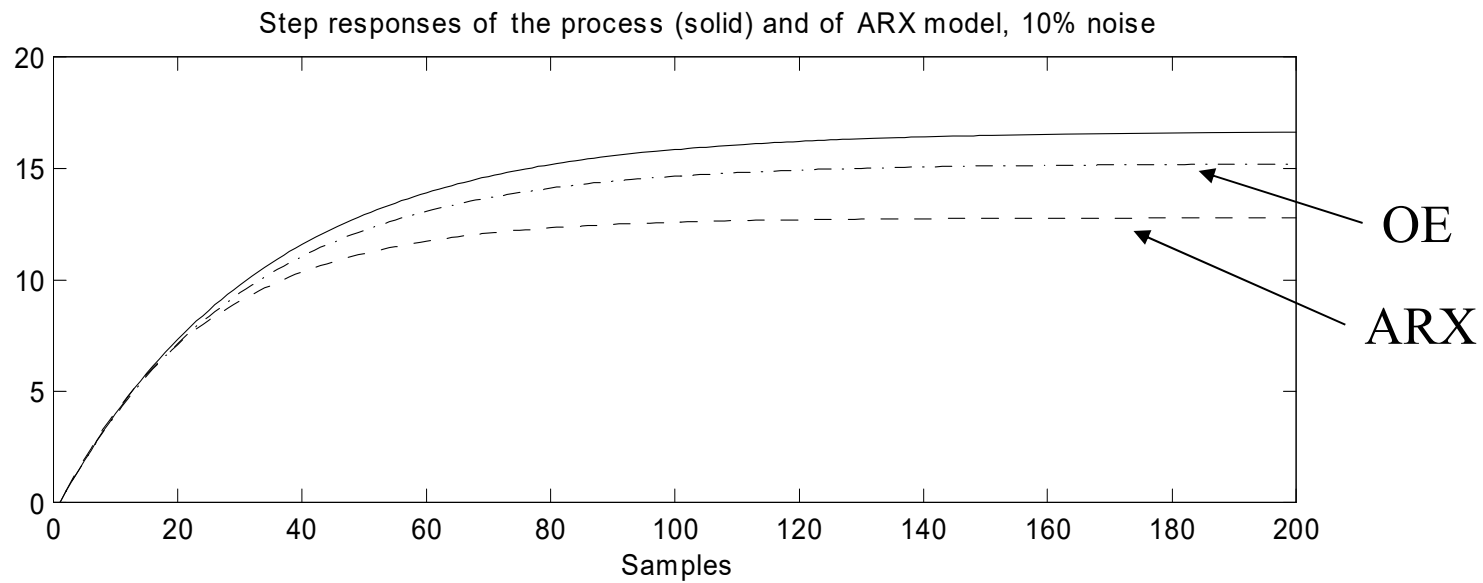
$$v(t) = \frac{1}{1 - 0.9q^{-1}} e(t)$$

$$\text{var}(v(t)) / \text{var}(y_o(t)) = 10\%$$

**Input: GBN, Tsw = 20 samples,
Number of data N = 1000**

5.2 Output-error method

Simulation Example



5.2 Output-error method

Comments

- Choose the error (residual) at the right location
- A good model structure may have a high cost in numerical optimization (long computation, local minimum, ...)
- The output error model will be biased for closed-loop data.

5.4 Prediction error methods

To achieve: 1) Consistency (unbiasedness) in closed-loop test
2) Efficiency (minimum variance) in both open loop and closed-loop.

5.4.3 ARMAX model

The true process is assumed to be (Åström and Bolin, 1965)

$$A^o(q)y(t) = B^o(q)u(t) + C^o(q)e(t)$$

where

$$A^o(q) = 1 + a_1^o q^{-1} + \dots + a_n^o q^{-n}$$

$$B^o(q) = b_1^o q^{-1} + \dots + b_n^o q^{-n}$$

$$C^o(q) = 1 + c_1^o q^{-1} + \dots + c_n^o q^{-n}$$

and $e(t)$ is white noise with zero mean and variance R

5.4.3 ARMAX model

The predictor is

$$\hat{y}(t | \theta) = \frac{B(q)}{C(q)}u(t) + \left[1 - \frac{A(q)}{C(q)}\right]y(t)$$

Introduce the prediction error

$$\varepsilon(t) = y(t) - \hat{y}(t | \theta) = \frac{1}{C(q)}[A(q)y(t) - B(q)u(t)]$$

The parameters of this model are determined by minimizing

$$V_{ARMAX} = \frac{1}{N} \sum_{t=1}^N \varepsilon^2(t) = \frac{1}{N} \sum_{t=1}^N \left(\frac{1}{C(q)}[A(q)y(t) - B(q)u(t)] \right)^2$$

Remarks:

- Good model structure for some control design method.
- Numerical optimization is necessary and no global minimum can be guaranteed.

5.4.4 Box-Jenkins model(method)

The true process is assumed to be

$$y(t) = \frac{B^o(q)}{A^o(q)} u(t) + \frac{C^o(q)}{D^o(q)} e(t)$$

where $e(t)$ is the unknown white noise sequence and

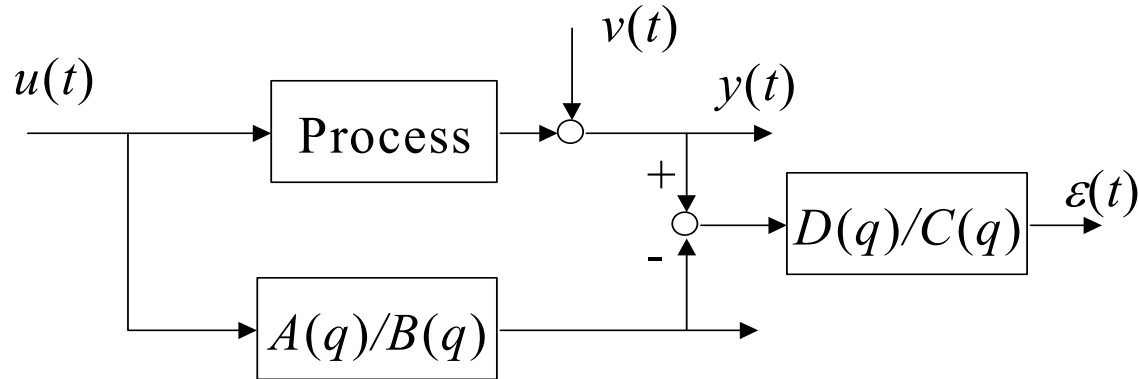
$$A^o(q) = 1 + a_1^o q^{-1} + \cdots + a_n^o q^{-n}$$

$$B^o(q) = b_1^o q^{-1} + \cdots + b_n^o q^{-n}$$

$$C^o(q) = 1 + c_1^o q^{-1} + \cdots + c_n^o q^{-n}$$

$$D^o(q) = 1 + d_1^o q^{-1} + \cdots + d_n^o q^{-n}$$

5.4.4 Box-Jenkins model (method)



The predictor is

$$\hat{y}(t | \theta) = \frac{D(q)B(q)}{C(q)A(q)}u(t) + \left[1 - \frac{D(q)}{C(q)}\right]y(t)$$

Introduce the prediction error

$$\varepsilon(t) = y(t) - \hat{y}(t | \theta) = \frac{D(q)}{C(q)} \left[y(t) - \frac{B(q)}{A(q)}u(t) \right]$$

The parameters of the model are determined by minimizing

$$V_{BJ} = \frac{1}{N} \sum_{t=1}^N \varepsilon(t)^2 = \frac{1}{N} \sum_{t=1}^N \left(\frac{D(q)}{C(q)} \left[y(t) - \frac{B(q)}{A(q)}u(t) \right] \right)^2$$

5.4.6 Properties of prediction error methods

Asymptotic distribution of parameter estimates

Assume: (1) The process is stable; (2) the model structure is correct; (3) the test signal is persistent exciting; (4) the minimization converges to its global minimum

Then we have (Ljung, 1999)

$$\hat{\theta}_N \rightarrow \theta^o, \quad \text{w.p. 1 as } N \rightarrow \infty$$

$$\sqrt{N}(\hat{\theta}_N - \theta^o) \in As N(0, P_\theta)$$

Averaging effect

$$\text{cov } \hat{\theta}_N \approx \frac{1}{N} P_\theta$$

The lower bound of P_θ is called Cramer-Rao bound.

5.6 Order selection

The purpose: To find the model order so that the model is most accurate for its use (in control).

Rough order selection methods: checking the rank of covariance matrices, plotting the singular value of Hankel matrix. Not accurate for noisy data.

Accurate approach: To first estimate models with increasing orders, and then to choose the best order using an error criterion.

Two issues: 1) What criterion? 2) What dataset?

5.6 Order selection

- In identification literature, the prediction error criterion is often used in model order selection.
- The use of a fresh data set for order selection is a very natural, the so-called cross-validation.
- The disadvantage of cross-validation is its high cost. It is more economical to use estimation data for order selection.
- The error criterion using the estimation data is a biased estimate of the error criterion using the validation data.
- Denote prediction error criterion using a validation data set as

$$V_{PE}^V = \frac{1}{N} \sum_{t=1}^N \hat{H}^{-1}(q) [y^v(t) - \hat{G}(q)u^v(t)]^2$$

- Denote prediction error criterion using the estimation data set as

$$V_{PE}^E = \frac{1}{N} \sum_{t=1}^N \hat{H}^{-1}(q) [y(t) - \hat{G}(q)u(t)]^2$$

5.6 Order selection

Assume that the white noise $e(t)$ is Gaussian and the model order is correct, then Akaike (1969) has derived an asymptotically unbiased estimate of V_{PE}^V using V_{PE}^E

$$EV_{PE}^V = \frac{N + d}{N - d} V_{PE}^E$$

This is called *final prediction error* criterion (FPE).

Here the factor $\frac{N+d}{N-d}$ is used to correct for the overfit effect or to punish the higher order.

5.7 Model validation

Purpose: To check if the obtained model is good enough for its use (control) and, *if not, provide remedy (what to do)*

- 1) Compare model property against process knowledge.
Rough estimate of the process gain, the dominant time constant.
Use model step response plots.
- 2) Check the Agreement of Different Methods.
Estimate FIR, ARX, ARMAX and Box-Jenkins models and compare their step responses and frequency responses.
- 3) Simulation

$$\begin{aligned}\hat{e}(t) &= y(t) - \hat{y}(t) \\ &= [G^o(q) - \hat{G}(q)]u(t) + v(t)\end{aligned}$$

Error small (<10%), OK for SISO, not for MIMO

5.7 Model validation

Residual analysis

1) $e(t)$ is white noise with zero mean. This implies the autocorrelation

$$\hat{R}_{\varepsilon}(\tau) = \frac{1}{N} \sum_{t=1}^N \varepsilon(t) \varepsilon(t - \tau)$$

should look like an impulse.

2) $e(t)$ and $u(t-1), u(t-2), \dots$, are uncorrelated.

$$\hat{R}_{\varepsilon u}(\tau) = \frac{1}{N} \sum_{t=1}^N \varepsilon(t) u(t - \tau) \approx 0, \quad \tau = 1, 2, \dots$$

Remark: Some thresholds are necessary to check the functions.

5.7 Model validation

A critical comment: Identification for control is not solved.

- The first three methods are more or less qualitative tests.
- The residual analysis is to check if the model agrees sufficiently well with the test data.
- None of the methods can tell if the model is suitable for control.
- If a model has passed the residual analysis, it only tells good model is obtained for the given data. But is the data is good enough (for control)?